WONG JIUN YAN

Electrical Engineering Student at NTHU | Analog & Mixed-Signal Circuit Design | Semiconductor Devices
Taiwan | wongjiunyan@gmail.com | linkedin.com/in/wong-jiun-yan

EDUCATION

National Tsing Hua University (NTHU) | Hsinchu, Taiwan | 09/2023 – 06/2027 Expected
Bachelor of Science in Electrical Engineering

Tsun Jin High School (循人中学) | Kuala Lumpur, Malaysia | 01/2017 – 11/2022
Science Stream | Ranked 2nd in class for 3 consecutive years; SPM 6As; School Merit Award for Science Special Study

SKILLS & CERTIFICATIONS

- EDA & VLSI Tools: Cadence Virtuoso, Synopsys Laker, HSPICE, Calibre DRC/LVS
- Core: Analog/Mixed-Signal IC Design, Power Electronics, Semiconductor Device Physics, Logic Design, Verilog HDL
- Programming: Python, Streamlit, C/C++ | Languages: Mandarin, English, Malay
- Certifications: Verilog HDL Design, Career Soft Skills Workshop (NTHU)

RESEARCH EXPERIENCE

NTHU Power Electronics & IC Design Laboratory | Hsinchu, Taiwan | 02/2026 – Present
Undergraduate Research Assistant

- Developing isolated gate driver architectures for high-frequency GaN power systems to improve switching reliability, power efficiency, and system-level robustness.
- Designed primary-side and secondary-side analog blocks, including frequency modulation, rectifier, and protection circuits, using Cadence Virtuoso.
- Investigated signal and power transmission through micro-transformer structures for galvanic isolation in high-voltage applications.
- Performed schematic design, custom layout, DRC/LVS verification, and post-layout analysis across process corners.

PROJECTS

Full-Custom 128 × 16 ROM Macro Design | 09/2025 – 12/2025

- Executed a full-custom design-to-verification flow for a 128 × 16 ROM macro with active power gating for high-speed and low-power operation.
- Optimized custom layouts in Synopsys Laker, achieving an area of 162,205 μm² and verified operation up to 136.8 MHz across 5 process corners using HSPICE.
- Completed schematic simulation, layout implementation, DRC/LVS verification, and pre-/post-layout performance analysis.

Two-Stage CMOS Operational Amplifier with Miller Compensation | 09/2025 – 12/2025

- Designed a two-stage CMOS operational amplifier with a PMOS differential input pair, active load, and Miller compensation for stable closed-loop operation.
- Achieved 61.9 dB DC gain, 10.68 MHz unity-gain bandwidth, 86° phase margin, and 28.9 μW total power consumption under HSPICE verification.

NTHU COPILOT: Agent-Based Course Planning Assistant | 03/2026 – 05/2026

- Built a tool-using AI agent system with Python and Streamlit to support course planning, schedule conflict checking, and structured academic decision-making.
- Integrated LLM-based natural language understanding with deterministic Python tools and safety guardrails to reduce hallucination risk.

LEADERSHIP & ADMINISTRATIVE EXPERIENCE

Student Assistant Roles, NTHU | Hsinchu, Taiwan | 01/2024 – Present

- Supported administrative operations at the Office of Global Affairs and Cashier Office, including document processing, data organization, and Excel-based record management.
- Assisted Animal Facility operations by following safety protocols and supporting reliable lab resource management.

NTHU Student Associations | Hsinchu, Taiwan | 09/2024 – 04/2025

- IT Officer, Malaysian Student Association: maintained digital platforms, organized internal records, and supported committee communication.
- Event Assistant, Overseas Compatriot Student Association: coordinated logistics and booth operations for student events, including the International Festival.